**A Project Report on**

# "Analyze and Forecast Dynamic Prices of Vegetables"

**Submitted in partial fulfillment of the requirements of the degree of Bachelor of Engineering (B.E.) (Information Technology)**

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# CERTIFICATE

This is to certify that the project entitled

**Analyze and Forecast Dynamic Prices of Vegetables**

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In partial fulfillment of degree of **B.E.** in **Information Technology** for termwork of the **Semester VII - Major Project- I** is approved.

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# Declaration

I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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# Abstract

The modern agricultural landscape faces a myriad of challenges, ranging from climate variability to market dynamics, all of which intricately influence the supply and demand of essential commodities like vegetables. In response to this complex scenario, this project presents research aimed at deciphering the influences on vegetable supply and demand using state-of-the-art of advanced computer algorithms.

As agriculture intersects with technology, the project stands at the intersection of data science, machine learning, and agri-economics. Drawing upon a comprehensive dataset that encompasses climatic variables, market trends, weather patterns, increased demand, and more, this project employs advanced algorithms to discern the intricate relationships between these factors and their impact on the vegetable ecosystem. The project's primary objective is to provide predictive insights into the fluctuations of vegetable pricing, enabling informed decision-making across the agricultural value chain. This holistic approach allows stakeholders to adapt to dynamic conditions and make strategic choices in response to the multifaceted challenges that affect the agricultural sector.

This research initiative not only addresses the practical implications of crop production and pricing but also contributes to the scientific understanding of the interplay between various influences on vegetable supply and demand. By leveraging the capability to process and analyze vast volumes of data, this research aims to unravel patterns and correlations that might otherwise remain hidden within the complexity of the agricultural domain.

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# Chapter 1 Introduction

## 1.1 Background

Agriculture is not just a fundamental source of sustenance; it is the cornerstone of human civilization. It forms the backbone of the global economy, directly affecting the lives and livelihoods of billions of people. The pricing, demand and supply of crops and vegetables have always been subject to a variety of factors, from climatic conditions to market dynamics.

Tomatoes are not just a fundamental source of sustenance; they are the cornerstone of human civilization. They form the backbone of the global economy, directly affecting the lives and livelihoods of billions of people. The pricing, demand, and supply of tomatoes have always been subject to a variety of factors, from climatic conditions to market dynamics.

However, in recent times, the world has witnessed an alarming increase in the frequency and intensity of price fluctuations in the tomato sector. These fluctuations have far-reaching consequences, rippling through the global economy and significantly impacting the lifestyle of people in both rural and urban areas. This delves into the multifaceted dimensions of how the prices of tomatoes are continuously fluctuating. Tomatoes are deeply ingrained in the history and culture of every society. For millennia, humans have relied on the cultivation of tomatoes for sustenance, trade, and progress.

The tomato sector is not just about growing food; it is a major driver of economic development, contributing significantly to a nation's GDP and providing employment for a substantial portion of the population. As such, it is often said that the state of the tomato industry is a reflection of the state of a nation.

Price fluctuations in the tomato sector can be attributed to a complex interplay of factors. These include natural variables such as weather patterns, pests, and diseases, which can decimate tomato harvests and cause shortages. Economic factors, such as changes in currency exchange rates, trade policies, and global market trends, can also drive price volatility. Furthermore, shifts in consumer preferences, technological

advancements, and governmental interventions play significant roles in shaping the prices of tomatoes. Machine learning offers a potent solution to mitigate price fluctuations in the tomato market. By analyzing historical data, ML models predict price movements, accounting for variables like weather, diseases, and market trends. ML can assess risk, providing insights into potential threats such as adverse weather events. Market sentiment analysis tracks public perception's impact on prices.

Optimizing the supply chain through ML minimizes overheads, while smart farming improves yields and crop health. Dynamic pricing strategies adjust to real-time market conditions, reducing price instability. Government interventions can also benefit from ML insights. These ML-driven strategies enhance price stability, ensuring a more resilient tomato market.

## 1.2 Motivation

Fluctuating crop and vegetable prices have substantial economic repercussions at the macro and micro levels. At the macroeconomic level, they can disrupt the stability of a nation's economy. Sharp spikes in food prices can contribute to inflation, leading to higher living costs for the population. Such inflation can erode the purchasing power of consumers, making it harder for them to meet their basic needs and invest in other sectors. Conversely, prolonged periods of low agricultural prices can reduce farmers' incomes, pushing rural communities into poverty. Food security, the ability of individuals and communities to ac- cess safe and nutritious food at all times, is a primary concern when it comes to fluctuating crop prices. For vulnerable populations, such as those in low-income countries or conflict zones, even small changes in food prices can have dire conse- quences. A sudden increase in the price of staple crops can lead to food shortages, malnutrition, and even famine. It's imperative to recognize that food security is not just about the quantity of food but also its affordability. The effects of fluc- tuating prices are particularly poignant for farmers. Farmers are at the forefront of agricultural production and are directly exposed to market volatility. When prices are high, farmers can benefit from increased incomes, but they are also vulnerable to the high costs of production and market fluctuations. On the other hand, when prices are low, farmers often struggle to cover their costs and may be forced to sell their produce at a loss, leading to financial distress and debt. Fluctuating crop prices can be a driving force behind rural-urban migration. When farmers experience prolonged periods of low agricultural prices and limited in-

come potential, they may seek better economic opportunities in urban areas. This migration can lead to overcrowding in cities, straining urban infrastructure and services. At the same time, rural areas may face a decline in agricultural production, impacting food security and the livelihoods of those who remain. Agricultural products are not confined to national borders. In today's globalized world, they are essential elements of international trade. Fluctuations in crop and vegetable prices can strain international trade relations, causing diplomatic tensions between countries. Disputes over trade agreements, tariffs, and import/- export quotas can arise as nations seek to protect their agricultural sectors or gain access to foreign markets. Intensive agriculture, driven by the pursuit of profitin the face of price fluctuations, can lead to environmental degradation. Overuse of pesticides, excessive irrigation, and mono-culture practices are often employed to maximize yields and profits. These practices can harm ecosystems, deplete water resources, and reduce biodiversity. Price-driven agricultural practices can thus contribute to ecological imbalances and climate change. To mitigate the impact of fluctuating crop and vegetable prices, communities and nations need to build resilience and adapt to changing circumstances. This includes diversifying agricultural practices, embracing sustainable farming techniques, and developing early warning systems to prepare for price volatility. In addition, governments and international organizations can play a significant role in supporting farmers and vulnerable populations during times of crisis.

## 1.3 Problem Definition

In the realm of agriculture and food supply chain management, the prices of vegetables are subject to constant fluctuations due to a wide array of influen-tial factors. These factors include the ever-changing climate conditions, seasonal variations, shifts in demand and supply, and the intricate dynamics of the market itself. The prices of vegetables can rise and fall unpredictably, often in response to the interplay of these variables. Accurate and timely analysis of these price vari- ations is of paramount importance for various stakeholders within the agriculture and food industry.

Farmers, for instance, depend on a sound understanding of price trends to make informed decisions about crop selection, planting schedules, and harvest timing. Distributors rely on pricing insights to determine when and where to source vegetables. Retailers need to anticipate price changes to maintain competitive pricing and manage inventory efficiently. Consumers benefit from stable and affordable prices, which are crucial for ensuring food security and affordability.

The primary objective of this project is to harness the capabilities of machine learning to address these challenges. By developing a machine learning-based solution, this project aims to offer a sophisticated tool that can effectively analyze historical and real-time data related to vegetable prices. By doing so, this project can provide not just data, but actionable insights. These insights empower stakeholders through- out the agriculture and food industry to make more informed and strategic decisions.

## 1.4 Scope

The project aims to create a machine learning-based solution for analyzing and predicting the dynamic prices of vegetables in the agriculture and food supply chain. This comprehensive endeavor will include the following aspects:

- The project will begin with the collection and preprocessing of historical and real-time data concerning vegetable prices. Data sources will encompass local markets, government records, and online databases. Careful data preprocessing will ensure data quality and consistency.
- To enhance the accuracy of price predictions, the project will involve feature engineering. A range of relevant features will be identified and constructed, such as weather conditions, seasonal patterns, economic indicators, and market trends. These features will be integrated into machine learning models.
- Multiple machine learning algorithms, including regression, time series anal- ysis, and ensemble methods, will be explored and employed to create predictive models for vegetable prices. These models will take into account historical data and heuristics.
- Real-time analysis capabilities will be developed to enable the continuous collection of new data and the generation of real-time price predictions. This will facilitate ongoing monitoring and adjustment of pricing strategies.
- The project will include the creation of interactive visualizations and dash- boards. These tools will present predicted prices and insights to end-users in a user-friendly format. The project will also encompass the interpretation of model outputs, shedding light on the critical factors influencing price fluctuations.
- A recommendation system will be devised to provide actionable strategies to stakeholders, such as farmers, distributors, and retailers. These recommendations will be rooted in the predicted price trends and may encompass guidance on planting or harvesting schedules, inventory management, and pricing strategies.

The project will conclude with the deployment of the machine learning solution in a production environment and integration with the systems of relevant stakeholders. This integration is vital to ensure the practical usability and effectiveness of the solution in the real-world context.

## 1.5 Limitations

The accuracy and availability of historical and real-time data on vegetable prices can present a significant hurdle. Variations in data quality and coverage, especially across different regions or sources, may influence the reliability of price predictions. The accuracy and availability of historical and real-time data on vegetable prices can present a significant hurdle.

Variations in data quality and coverage, especially across different regions or sources, may influence the reliability of price predictions. While machine learning models are powerful tools, they are not infallible. Model performance may fluctuate across different types of vegetables, regions, or time periods.

Maintaining consistent high accuracy across these variables can be a complex endeavor. The volatile nature of vegetable prices is influenced by numerous factors, some of which are challenging to predict accurately. External shocks, such as natural disasters or economic crises, can disrupt market dynamics, making predictions even more challenging.

Further this report provides the literature survey will be explored in which the topics like need of domain , survey of subroutine and tools, literature survey comparison will be seen .This report further explores the problem formulation which describes the aim, objectives and the system design which is going to be used in order to build the project.

# Chapter 2 Literature Review

The analysis and prediction of dynamic vegetable prices in the context of the agriculture and food supply chain have garnered significant attention in recent years. Researchers and practitioners alike have recognized the potential for data-driven approaches, particularly those harnessing machine learning, to revolutionize pricing strategies and enhance decision-making in this critical sector.

Several studies have explored the use of historical price data, market trends, and external variables as essential components in price prediction models. Heuristics that incorporate factors such as weather conditions, seasonal patterns, and economic indicators have shown promise in improving the accuracy of predictions. The significance of real-time analysis and prediction capabilities cannot be overstated. In a dynamic market, the ability to adapt to changing conditions swiftly is a competitive advantage.

Ethical considerations surrounding predictive pricing have become a pressing concern. Researchers have delved into the potential biases introduced by machine learning models and emphasized the importance of addressing fairness and transparency in pricing strategies.

The role of recommendation systems in guiding stakeholders has also been explored. Devising strategies based on predicted price trends has been a central theme in recent literature. Farmer-centric guidance on planting schedules, distributors' procurement decisions, and retailers' pricing strategies are areas of growing research, with potential applications.

Overall, the literature reflects a growing consensus on the potential of ma- chine learning to enhance pricing strategies within the agriculture and food supply chain. The reviewed studies emphasize the value of data quality, feature engineering, real-time analysis, and ethical considerations in this evolving landscape. The development of adaptable, scalable, and ethical machine learning solutions holds the potential to provide actionable insights, reduce food wastage, and enhance food security, benefitting a wide array of stakeholders within the industry.

## 2.1 Need of Domain

In the realm of developing machine learning solutions for complex real-world problems, the integration of domain expertise is indispensable. Domain expertise refers to in-depth knowledge and understanding of the specific industry or field in which a machine learning application is being developed. While machine learning algorithms can effectively analyze and process data, they often lack the contextual understanding required to make informed decisions and generate actionable insights.

In the context of agriculture and food supply chain management, domain expertise is vital for several reasons. It enables the identification of relevant variables, the formulation of appropriate features, and the selection of heuristics that are specific to the agricultural sector. Understanding the intricacies of crop cycles, market dynamics, climate conditions, and supply chain logistics allows for the creation of machine learning models that are not only accurate but also relevant and practical.

Moreover, domain expertise provides the necessary insight to interpret the results of machine learning models correctly. For example, it helps distinguish between a genuine pricing trend and a short-lived anomaly, preventing hasty decisions. It also aids in addressing ethical concerns, such as the potential for bias in algorithms, ensuring that the machine learning application respects the values and norms of the industry.

## 2.2 Survey of subroutines and tools

Following algorithms are used in analyzing dynamic prices of vegetables:

- Data Scraping with Selenium: You are using Selenium, a web automation tool, to extract daily commodity prices from websites. Selenium is particularly useful for websites that use JavaScript or require interaction to access data. It allows you to automate the data retrieval process efficiently.

- Data Preprocessing: After scraping the data, you'll need to preprocess it. This involves tasks such as handling missing values, cleaning the data, and ensuring data quality.
- Time Series Analysis with SARIMA (Seasonal Autoregressive Integrated Moving Average): SARIMA is a powerful time series forecasting method. It's used to model and predict time-dependent data, which is often the case

- with commodity prices. SARIMA models consider both autoregressive and moving average components, along with seasonality. This helps in capturing trends and patterns in the data.
- Regression Methods: Regression analysis is used to understand the relationships between variables. In your context, regression methods may be applied to identify factors that influence commodity prices. This could include building models to predict prices based on other variables like eco- nomic indicators, weather conditions, or supply and demand factors.

## [1] Vegetable Price Prediction using Data Mining Classification Techniques – 2019

The project titled "Vegetable Price Prediction using Data Mining Classification Techniques – 2019" is centered on forecasting the weekly and monthly market prices of tomatoes with the use of an AMM (Association for Mixed Models) model. The dataset utilized in the project covers only a three-year period. This limitation poses a challenge as a more extensive dataset would provide a broader historical context for price trends and fluctuations, potentially leading to more accurate predictions. The project acknowledges that Artificial Neural Networks (ANN) resulted in lower accuracy. This highlights the need for further investigation into alternative machine learning models or techniques that may be better suited for the specific task of tomato price prediction. The project is confined to the Coimbatore region. While this specificity can be beneficial for local stakeholders, it limits the generalizability of the model to other regions or markets. Expanding the project's scope to include a more diverse range of locations could enhance its practicality.

## [2] Economic Analysis of Profitability In Tomato Production At different seasons and market prices: A study in Kolar district of Karnataka 2018

The research paper titled "Economic Analysis of Profitability In Tomato Production At different seasons and market prices: A study in Kolar district of Karnataka 2018" delves into the economic viability of tomato cultivation, taking into account various

seasons and market price fluctuations within the Kolar district. To carry out this study, the authors engaged with 150 tomato farmers in the region and meticulously gathered data to assess the overall cost of cultivation, returns, and the resultant profitability of tomato farming.

First, the study is geographically limited to Kolar district, Karnataka. This means that the findings and conclusions drawn in the paper may not be readily applicable to other regions or districts with distinct agricultural dynamics. Agricultural practices, market conditions, and profitability may vary significantly from one area to another.

Second, the study focused primarily on the economic aspects of tomato production and profitability. It did not take into consideration the potential impact of pests and diseases on the profitability of tomato cultivation. Pests and diseases can substantially affect crop yields and quality, which, in turn, can have a substantial influence on a farmer's bottom line. An analysis of these factors would have provided a more comprehensive view of the challenges and opportunities in tomato farming.

In summary, while the paper provides valuable insights into the economics of tomato production in Kolar district, its findings may not be universally applicable due to its regional specificity. Additionally, the absence of an analysis on the impact of pests and diseases is a notable limitation, as these factors are critical in understanding the overall profitability and sustainability of tomato farming in any region.

## [3] Solving Onion Market Instability by Forecasting Onion Price Using Machine Learning Approach - 2020

The research paper titled "Solving Onion Market Instability by Forecasting Onion Price Using Machine Learning Approach 2020" focuses on the crucial task of predicting onion prices through the application of machine learning techniques. Price forecasting in the onion market is essential for enabling stakeholders to make informed and timely decisions, thereby adapting their strategies to the often volatile and ever-changing conditions in the market.

Firstly, it grapples with the unusual behavior of data. In the realm of machine learning and data analysis, anomalous or irregular data patterns can pose a significant challenge. Unusual data behavior might be characterized by extreme price fluctuations or unpredictable trends. Dealing with such data can necessitate advanced techniques and models that are capable of handling these idiosyncrasies.

Secondly, the paper mentions a low number of records. A limited dataset can hinder the predictive capabilities of machine learning models. In this context, a smaller number of records might result in models that lack the depth and breadth of historical data needed to make highly accurate and robust predictions. Expanding the dataset, if feasible, could potentially improve the model's forecasting accuracy.

In summary, this research paper plays a vital role in addressing the instability in the onion market by harnessing the power of machine learning for price forecasting. However, it encounters challenges related to peculiar data behavior and the restricted number of records, which can affect the accuracy and reliability of the price predictions. Mitigating these challenges and exploring techniques for handling unusual data patterns are essential steps in enhancing the paper's contributions to addressing onion market instability.

## [4] Modelling and Forecasting of Area, Production and Productivity of Tomatoes in Haryana and India - 2022

The paper titled "Modelling and Forecasting of Area, Production and Productivity of Tomatoes in Haryana and India 2022" delves into a comprehensive investigation of the long-term trends in tomato cultivation, encompassing factors such as the area under cultivation, total production, and productivity. The analysis extends from 1991 to 2018, shedding light on the dynamics of tomato farming in both Haryana and the broader context of India.

The authors of the paper employed a variety of modeling techniques, including linear, exponential, quadratic, S-curve trends, and ARIMA models. These methodologies were utilized to capture the historical data trends and make predictions about future developments in tomato farming.

First, the data analyzed in the paper spans from 1991 to 2018. While this dataset

provides valuable historical insights, it is worth noting that trends in agriculture are subject to change due to evolving factors such as technology, climate, market demands, and government policies. Therefore, the paper's findings might not fully represent the current or future state of tomato farming.

Second, the study is primarily focused on the state of Haryana and, to a broader extent, India. As such, the results may not be universally applicable to other regions or countries with distinct agricultural landscapes and influences. Tomato cultivation practices and trends can differ significantly from one place to another.

In summary, the paper offers a comprehensive analysis of tomato farming trends in Haryana and India, leveraging various modeling techniques to understand past and potential future developments. However, it's crucial to recognize that the study's findings are rooted in historical data and the specific context of Haryana and India, and that these insights may not be directly transferable to other regions or reflect more recent developments in tomato agriculture.

## 2.3 Literature Survey Comparison

| Sr no | | Author | Objective | Limitation |
|---|---|---|---|---|
| 1 | Vegetable Price Prediction using Data Mining Classification Techniques – 2019 | G. M. Nasira<br>N. Hema | Weekly and Monthly Market Price Prediction of Tomatoes using AMM Model. | Dataset has only data for 3 years<br>ANN has lower accuracy<br>Limited for Coimbatore.<br>No real time analysis of data |
| 2 | Economic Analysis Of Profitability In Tomato Production At different seasons and market prices: | Vanitha S.M.<br>Chinnappa Reddy B.V.<br>Gajanana T.M. | The paper investigates the economic profitability of tomato production in different seasons and market prices in Kolar district, | The study was conducted only in Kolar district, so the results may not be generalizable to other regions.<br>The study did not consider the impact of pests and diseases on the profitability of tomato production. |

| | | | | |
|---|---|---|---|---|
| | A study in Kolar district of Karnataka 2018 [2] | | Karnataka. The authors collected data from 150 tomato growers in the district and used it to analyze the cost of cultivation, returns, and profitability of tomato production. | |
| 3 | Solving Onion Market Instability by Forecasting Onion Price Using Machine Learning Approach 2020 [3] | M. M. Hasan<br>M. Tuz Zahara<br>M. M. Sykot<br>R. Hafiz<br>M. Saifuzzaman | The primary goal of the paper is to forecast onion prices using machine learning techniques. Forecasting prices can help stakeholders make informed decisions and adapt their strategies to changing market conditions. | The main limitation is unusual behavior of data and low number of records. |
| 4 | Modelling and Forecasting of Area, Production and Productivity of Tomatoes in Haryana and India 2022 [4] | Nimbrayan<br>Parveen & P K, Muhammed & Chandanshive<br>Aniket | The paper investigates the trends in the area, production, and productivity of tomato crop in Haryana and India from 1991 to 2018. The authors used linear trend, exponential trend, quadratic trend, S-curve trend, and ARIMA modelling techniques to model the data. | The data used in the study is from 1991 to 2018, so it is possible that the trends may change in the future.<br>The study only focuses on the state of Haryana and India, so the results may not be generalizable to other regions. |

Table 2.1. Literature Survey Comparison

# Chapter 3

# Problem Formulation

## 3.1 Aim and Objectives

### 3.1.1 Aim

This project's core objective is to create a data-driven solution for understanding and predicting the ever-changing prices of vegetables. Its primary aim is to offer valuable insights and tools to consumers, farmers, and industry players to help them make well-informed decisions in the volatile vegetable market. The project encompasses tasks like data collection and processing, exploratory data analysis, feature engineering, model selection and training, real-time data integration, clear visualization and reporting, user-friendly application development, and the implementation of alert and feedback mechanisms. Ongoing performance assessment, educational resources, scalability, cost-benefit analysis, and compliance with relevant regulations are integral aspects of the project to ensure its accuracy, usability, and alignment with pertinent requirements.

### 3.1.2 Objectives

Following objectives are listed based upon problem statement and gap identification:

- Real-time Price Forecasting: Implement a system that continuously updates price predictions in real-time or near-real-time as new data becomes available. This ensures that stakeholders have access to the most current pricing information.

- Anomaly Detection: Develop anomaly detection techniques to identify ab-normal price fluctuations or market irregularities. The system should alert users when significant anomalies occur, allowing for timely responses.

- Geographic and Seasonal Analysis: Analyze the data to determine how vegetable prices vary by region and season. Understand the impact of geo- graphical and climatic factors on pricing fluctuations.

- Visualization and Reporting: Create data visualizations, dashboards, and reports that present the analysis and insights in an easily understandable format. Effective communication of findings is crucial for decision-makers.
- Feature Engineering: Identify and select relevant features that influence vegetable prices, such as weather conditions, seasonal variations, demand factors, and economic indicators. Effective feature engineering ensures that the model considers all pertinent factors.

## 3.2 System Design

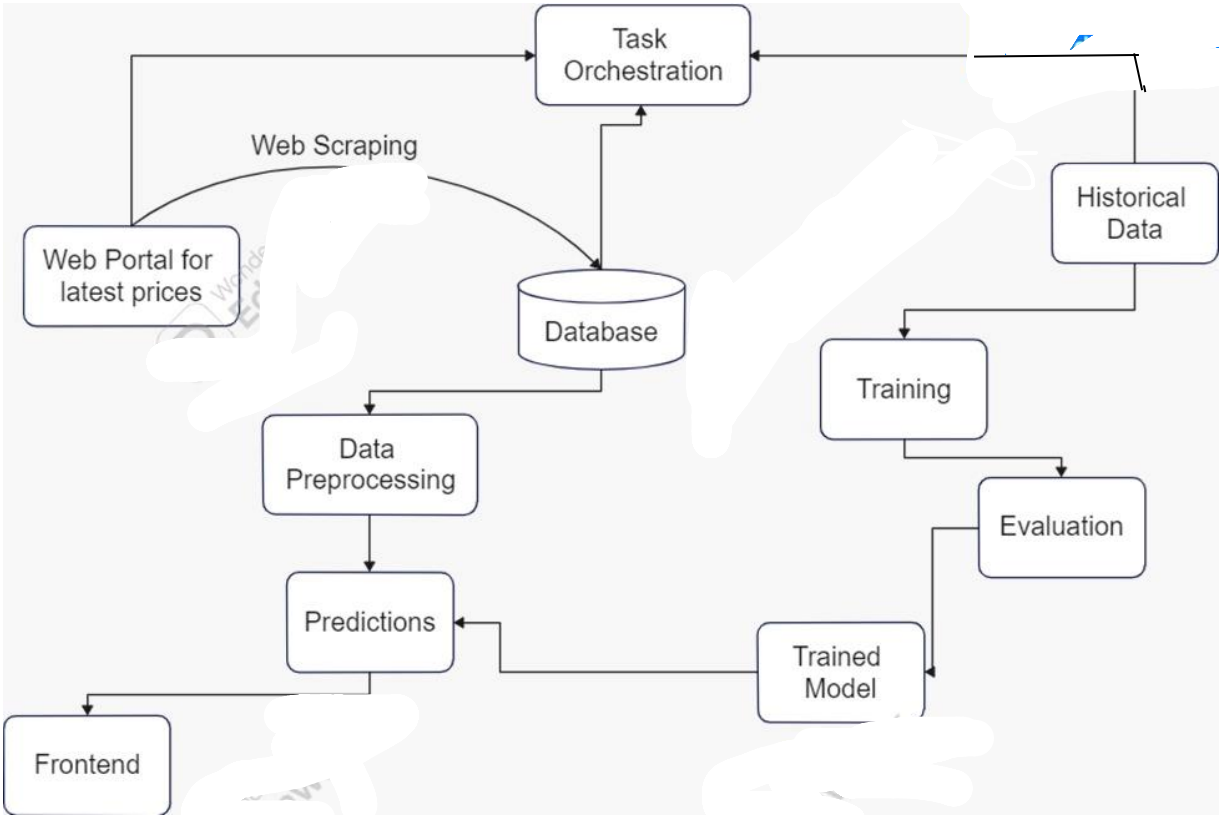
The process outlined involves a comprehensive workflow in the realm of data-driven decision-making, particularly in the context of tomato price prediction. It commences with the collection of historical data, wherein information regarding tomato prices is systematically extracted from websites through web scraping techniques. This initial dataset, though valuable, is often raw and unstructured, necessitating data preprocessing. During this phase, the data is cleaned, missing values are handled, and any anomalies or inconsistencies are addressed to ensure the dataset's quality.

Following data preprocessing, the core of the operation takes place with the training of a machine learning model. This model is designed to analyze the preprocessed historical data, discern patterns, and establish relationships within the dataset. The goal is to equip the model with the ability to make accurate predictions regarding tomato prices based on the various features provided in the data, such as time and location.

However, the efficacy of the model is not assumed; instead, it is rigorously evaluated. This evaluation involves assessing the model's performance using a distinct dataset that was not part of the training process. The aim is to gauge how well the model can generalize to new, unseen data and determine its reliability and accuracy in making predictions.

Finally, once the trained model successfully passes the evaluation phase, it is put into practical use. The model is deployed to forecast future tomato prices based on the input features provided. This forecasting capability has the potential to offer valuable insights to stakeholders within the tomato industry, ranging from farmers and distributors to policymakers and consumers. It can aid in making informed decisions, implementing pricing strategies, and managing risks associated with tomato price fluctuations. This holistic approach, encompassing data collection, preprocessing, model training, evaluation, and prediction, serves as a powerful tool in navigating the complexities of the tomato market.

### 3.2.1 Architectural Diagram

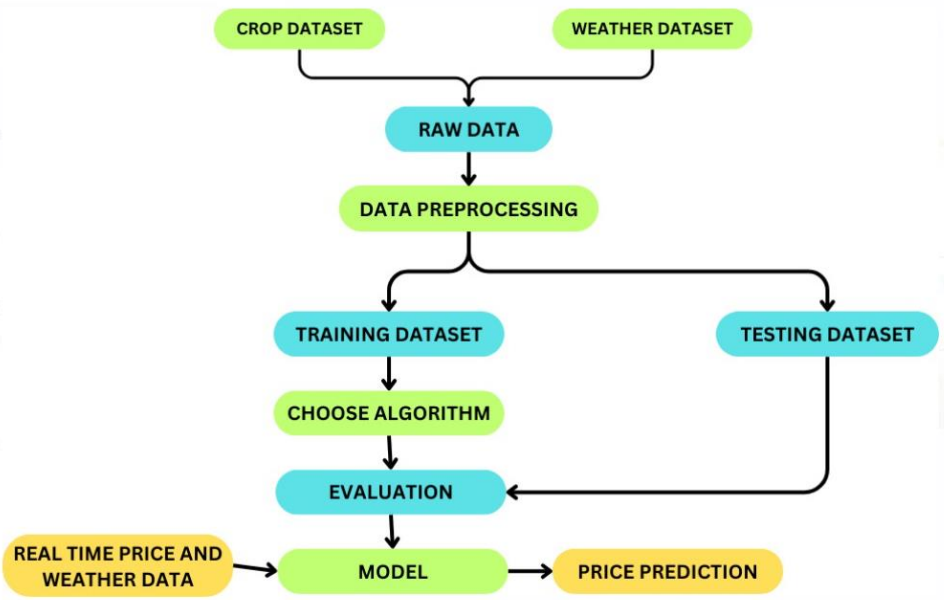



3.1 Architectural Diagram

The architectural diagram describes the working of this particular project. There are different components of the data that is present Historical data- This represents the historical data

which consists of the tomato prices. This particular dataset is used for the training purpose. Database- This consists of all the data which is available on the tomato along with the weather data. This also consists of the data which is continuously being scrapped by the web scrapper which will be used for the price prediction purposes. Web portal for latest prices- This refers to the website which is being used in order to check the daily prices of tomatoes. The historical data is taken and used for the model training. After the training the model is being evaluated and then the trained model is used for the prediction of the tomato prices. The trained model is provided after preprocessing the data that is scrapped from the website .

### 3.1.1 Flowcharts




3.2 Flowchart

First step is to collect the data from various sources in order to create a dataset. In the following project is using two types of data for the price prediction purposes that is the price of the vegetable and the weather data for growing that particular vegetable that is tomato in this case. After the collection of data these datasets needs to be merged together in order to form on single dataset which would consist the data of the prices as well as the weather together After the dataset is ready it needs to be preprocessed so that the data present in the dataset can be used for the training of the model for the price prediction. Then the training of the model takes place and then the model is evaluated. Hence this model is used for the price prediction of the tomatoes.

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